Supplementary Appendix to

Association of Logic's Hip Hop Song *1-800-273-8255* with Lifeline Calls and Suicides in the United States: Interrupted Time-Series Analysis

Thomas Niederkrotenthaler, Ulrich S. Tran, Madelyn Gould, Mark Sinyor, Steven Sumner, Markus J. Strauss, Martin Voracek, Benedikt Till, Sean Murphy, Frances Gonzalez, Matthew J. Spittal, John Draper

Table of Contents

Supplementary Text S1. The Statistical Model.....2

Supplementary Text S2. Identification of Events and Durations of Impact.....5

Supplementary Table S1. Identification of Suicides by Celebrities..... 7

Supplementary Table S2. Analysis of the Lifeline Time Series, Adjusted for all of the Identified Exogenous Events..... 9

Supplementary Table S3. Analysis of the Suicide Time Series, Adjusted for all of the Identified Exogenous Events.....11

Supplementary Table S4. Sensitivity Analysis of the Lifeline Calls Time Series, Adjusted for all of the Identified Exogenous Events: Unique calls.....13

Supplementary Table S5. Sensitivity Analyses of the Lifeline and the Suicide Time Series: Associations with All Logic Main Events, Using a Pre-Intervention Cut-Off Date of April 27, 2017, and Adjusted for all of the Identified Exogenous Events.....15

Supplementary Table S6. Sensitivity Analyses of the Lifeline and the Suicide Time Series: Associations with All 5 Target Events, Adjusted for all Identified Exogenous Events.....16

Supplemental Table S7. Annotated SPSS Syntax of the Statistical Models.....17

Supplemental Text S3. Characteristics of Lifeline users: location, gender, age, primary presenting problem during call.....20

Supplemental Figure S1. Daily number of (A) Tweets including Logic and 1-800-273-8255, (B) calls to the National Suicide Prevention Lifeline, and (C) suicides -- United States, April 2017 to April 2018.....23

Supplementary Text S1. The Statistical Model

SARIMA (seasonal autoregressive integrated moving average) modeling was used to analyze the daily time-series data. SARIMA modeling allows for a flexible approach to the modeling of dependent time series, of outliers therein, and of independent (time-varying) variables that might be associated with changes in the dependent time series. All of this was relevant in the present study, as we investigated the temporal associations between discrete events related to Logic's song and Lifeline calls and suicides, which themselves were affected by background events, but also by outliers and technical anomalies related to the registration of calls. For an introduction to this class of models and their application in epidemiological research, see Schaffer et al. (2021).

SARIMA models are characterized by six parameters, SARIMA(p,d,q)(P,D,Q). p denotes the position or number of time lags which autoregressively affect current values, d denotes the number of times the differences between consecutive values were computed to remove trends and to reduce non-stationarity in the time series, and q denotes the position or number of current and past error terms which affect current values (random shocks); P, D, and Q are the respective parameters of a SARIMA model with a periodicity of 7 (i.e., conforming to the seven days of the week). In the present study, seasonality therefore related to weekly periodicities in the data.

Stationarity is an important requirement of SARIMA modeling. This requires that the mean (and the variance/autocovariance) of the time series remain constant over time. Stationarity demands the removal of time trends, which was accomplished via the differencing operation in the SARIMA model (i.e., the integration part of the model) for the Lifeline call time series (see below). A very slight visually observable linear trend in the suicide time series was modelled via a discrete level shift on August 13, 2014 that was identified by the SPSS Expert Modeler function (see below). The R^2 for this linear trend (in an ordinary least squares regression model) in the full suicide time series data amounted to 24%. This value decreased to 4% and 5% in the segments of the same time series before and after August 13, 2014, respectively. This indicates that a single level shift at a single point of the time series reduced non-stationarity by a meaningful amount. No seasonal (in our case, weekly) trends were observed in the present time series, which would have required seasonal differencing.

The selection of models was aided by the SPSS Expert Modeler function, version 26 (IBM), choosing models with the lowest Bayesian information criterion value, highest stationary R^2 value (the variance attributable to the fitted time-series model), and, where possible, a not significant Ljung-Box Q statistic (indicating whether residuals could be assumed white noise, with stated df). The Q statistic tests whether the autocorrelations of the model residuals at any or some lags are different from zero. A violation of this assumption indicates a relative loss of efficiency of parameter estimation. However, the Q statistic is also inflated by heteroscedastic data structures (see Hatemi-J, 2004, for the multivariate case). Thus, we accepted a significant Ljung-Box Q statistic for the slightly heteroscedastic Lifeline time series. The inclusion of further autoregressive or moving average terms would not have resulted in a non-significant Q statistic. As heteroscedasticity was only slight, and would have mainly affected time-series prediction (which was not part of this analysis), we did not apply data transformations (such as a log-transformation) on this time series, as this would have rendered the interpretation of parameter estimates unnecessarily difficult.

We visually inspected the time-series plots for trends in the data, and the plots of the autocorrelation function (ACF) and the partial autocorrelation function (PACF) for relevant autocorrelative patterns, in addition to the *Q* statistic. Due to the many outliers, model building progressed manually for the Lifeline data, first performing a differencing operation ($d = 1$) to remove a clearly visible trend in the time-series plot and then adding autoregressive parameters and random shocks one at a time (starting with lag 1 in both cases), until any additional parameters did not reach significance. Outliers were subsequently detected with the SPSS Expert Modeler function and incorporated into the final model.

Concerning model selection, first and foremost, we deliberately aimed at obtaining the simplest and most parsimonious models that were compatible with the data (following the principle of Occam's razor); i.e., models with the smallest possible orders of the parameters *p*, *d*, *q*, *P*, *D*, and *Q*. This approach conforms to recommendations in the literature, which highlight that there may not be only one "right" model and that the most parsimonious model should be selected (e.g., Schaffer et al., 2021).

The associations of the investigated events with the dependent time series were modelled as discrete pulses, i.e., sudden changes from the baseline, which start and end with the identified events (see Schaffer et al., 2021). This was done because we wanted to model any associations with calls and suicides during periods of strong public attention to the specific events identified (see Supplemental Text S2 for the identification of events and their duration). We assumed that any associations during these periods would be reflected in immediate associations with calls and suicides. The approach is consistent with studies indicating that public attention to (social) media discourse is short-lived, with e.g., lifespans of a tweet normally not extending to more than a few hours (Bray, 2012). Thus, we assumed that if there were no immediate associations between promoting the narrative of hope and the Lifeline number, as assessed in the present study, any possible associations later on would be highly unlikely.

SARIMA modelling (and time-series modelling in general) does not allow us to directly establish evidence of causal effects. For instance, outcomes still could have been caused by variables which were not part of the investigated model. However, in the course of "natural experiments" and in the absence of the possibility of conducting randomized controlled trials, time-series modelling rigorously allows controlling for pre-event and post-event trends (both in the short term and the long term) and, for this reason, provides a more robust and valid assessment of the associations of observed events with outcomes than other approaches (see also Schaffer et al., 2021). Therefore, our analysis provides evidence of association, controlling for temporal effects, but also potentially important exogenous side events which happened during the time period of interest.

In the following, Supplemental Text S2 provides details about the identification of Logic-related media events and durations of impact. Supplemental Table S1 provides information relevant to the identification of suicides of celebrities during the period April 7, 2017 and December 31, 2018.

References

Bray P. *When is my Tweet's prime of life? (A brief statistical interlude.)* Published November 12, 2012. Accessed June 24, 2021. https://moz.com/blog/when-is-my-tweets-prime-of-life

Hatemi-JA. Multivariate tests for autocorrelation in the stable and unstable VAR models. *Economic Modeling*. 2004;*21*:661-683.

Schaffer AL, Dobbins TA, & Pearson SA. Interrupted time series analysis using autoregressive integrated moving average (ARIMA) models: A guide for evaluating large-scale health interventions. *BMC Medical Research Methodology. 2021; 21*: 58. https://doi.org/10.1186/s12874-021-01235-8

Supplementary Text S2. Identification of Events and Durations of Impact

Based on our collaboration with the Lifeline and as outlined in the preregistration (AsPredicted #62469, titled 'Logic song', on April 2, 2021), we were primarily interested in associations of calls and suicides with the three main events related to Logic song: (A) The song release (April 28, 2017); (B) Logic's performance at the MTV Video Music Awards with (August 27, 2017), and (C) his performance at the Grammy Awards (January 28, 2018). These three events were widely acknowledged by the media and the National Suicide Prevention Lifeline registered increases in call volumes after (A) and (B), without formally researching these increases.

In accordance with previous research about media events and their association with suicides, we used social media, i.e., Twitter volumes, to assess any further periods of strong public attention.[1-2] We downloaded daily numbers of all tweets geolocated in the United States using the search terms "Logic" and "1-800-273-8255" between April 1, 2017 and March 30, 2018 in order to capture a full time series of tweets that might potentially indicate further events of interest.

A visual inspection of the resulting daily time series clearly indicated a strong uptick in tweets on this topic starting at the MTV performance, a second peak at the Grammy Awards, and the third-largest peak at song release (see Figure 1). We then screened a random sample of Tweets, as obtained from Brandwatch (www.brandwatch.com) to check whether the content of tweets at the respective peaks were indeed associated with the specific events and all were. The bimodal peak for the MTV Music Awards turned out to be related to a first large peak about his performance at the MTV Awards, and a second smaller peak related additionally to a further performance at the Ellen DeGeneres Show (a talk show on the CBS network), as well as a South Park parody of the song (https://www.youtube.com/watch?v=e1HrwF9l-c4).
The visual inspection showed that Tweets about the song remained elevated between these two peaks; we therefore coded the entire period as impact period.

Two additional peaks were identified by visual inspection. We did a search of random tweets during the respective periods, which revealed that peak (D) was related to the video release, whereas peak (E) was related to a CBS network report about increases in calls to the Lifeline related to the song, as communicated by the Lifeline. There were no other noteworthy peaks in the time series.

In order to assess the durations of most likely impact of each of the peaks identified, we visually inspected the time series anew. We marked all days starting with the respective event and included each day until visual inspection suggested a return close to baseline. For events (A), (B), (C), and (D), this resulted in the estimated durations of impact as shown in Table 1. For event (E), visual inspection suggested an increase for 6 days starting with October 10, 2017, but, as outlined in the next paragraph, we ultimately used a 3-day period for this event.

To more fully assess whether the decisions from this qualitative approach of visual inspection were consistent with formal quantitative assessment, we performed a post-hoc change-point analysis.[3] Specifically, we used R version 4.1.1. (August 10, 2018) and the change-point package (https://cran.r-project.org/web/packages/changepoint/changepoint.pdf) to perform a pruned exact linear time

(PELT) method for the investigation of changepoints in the time series of tweets to assess the volatility in the time series.[4] Changepoints were identified via searching for points of high volatility in the time series, using the "cpt.var" command in the change-point package. We set no restrictions on the number of change points and used the default settings of the procedure. The results of this analysis were entirely consistent with the findings from visual inspection for events (A), (C), and (D). For (B), the procedure identified a total of 4 change-points, which were consistent with the start and end dates of the two peaks related to the MTV Video Music Awards previously already identified visually. With regard to event (E), the model suggested increased volatility for three days, which slightly differed from the 6-day period identified visually. We decided to use this 3-day period for modelling. A sensitivity analysis using the 6-day period for event (E) suggested that results were largely unchanged.

The specific changepoints identified were: April 27, 2017; April 29, 2017; August 16, 2017; August 20, 2017; August 27, 2017; September 3, 2017; September 20, 2017; September 23, 2017; October 9, 2017; October 11, 2017; January 28, 2018; and January 30, 2018.

Note that these dates mark dates immediately before the respective change; i.e., the visually identified dates all had one day added.

References

1. Niederkrotenthaler T, Stack S, Till B, et al. Association of increased youth suicides in the United States with the release of 13 Reasons Why. *JAMA Psychiatry*. 2019;76:933-40. doi:10.1001/jamapsychiatry.2019.0922
2. Niederkrotenthaler T, Till B, Garcia D. Celebrity suicide on Twitter: activity, content and network analysis related to the death of Swedish DJ Tim Bergling alias Avicii. *J Affect Disord*. 2019;245:848-855. doi:10.1016/j.jad.2018.11.071
3. Chen J, Gupta AK. Parametric statistical change point analysis. New York, NY: Birkhäuser Basel; 2000.
4. Killick R, Fearnhead P, Eckley IA. Optimal detection of changepoints with a linear computational Cost. *J Am Stat Assoc*. 2012;107:1590-8.

Supplementary Table S1. Identification of Suicides by Celebrities

We searched the Wikipedia entries on suicides by celebrities in the 21st century (see https://en.wikipedia.org/wiki/List_of_suicides_in_the_21st_century) to identify American and international celebrities who died in the period of April 7, 2017 (start of the post-period) until December 31, 2018 (the end of the observation period). Persons widely reported to be convicted of severe crimes (e.g., homicide) were excluded because of evidence that suicides by individuals portrayed as criminal are not associated with subsequent increases in suicides.[1] All of the identified names are listed below, along with their occupation and date of death. In order to identify those who received considerable public attention, and would require inclusion in the models, we used Brandwatch to identify the number of tweets in the month after the respective death. To do so, we searched for the name of the individual, using a similar approach as outlined in the main paper for the identification of tweets related to the Logic song. We also did a basic check of tweet content to confirm that tweets were indeed about the specific suicide death. 50,000 tweets were defined as a cutoff to define strong public attention. This threshold was reached by 6 individuals, with 5 of them each reaching more than 300,000 tweets in the first month after their death. As indicated by the content check of tweets, the cause of death by one of the 6 individuals, Verne Troyer, was originally not reported as a suicide death until much later, when there was no strong public attention any longer. Because the large wave of tweets on his death did neither contain any cause of death, nor any speculations about his death being a suicide, we did not include this suicide in the model. For the remaining 5 individuals, we checked for the duration of public attention by analyzing the number of tweets in the second month (i.e., starting with day 31 after the death) after the suicide death. For two of them (Chris Cornell and Avicii aka Tim Bergling), the volume had already returned to a very low level after one month. For these celebrities, we assumed a 30-day period as possible duration of impact. For three celebrities (Kate Spade, Anthony Bourdain, and Chester Bennington), the mean number of tweets per day was still more than 1,000 tweets per day in the second month, and levelled off considerably by the end of that month. We assumed a duration of 60 days for the possible duration of impact for these celebrities. This was also consistent with specific research investigating the associations of Spade's and Bourdain's suicides with suicides in the United States.[2] In general, the approach to use a period of 1-2 months depending on the celebrity is consistent with the literature.[3]

Name	**Occupation, nationality**	**Date of death**	**Number of Tweets, first month after death**
Chester Bennington	American singer and songwriter	July 20, 2017	465,184
Avicii (Tim Bergling)	Swedish DJ, remixer, record producer, musician, and songwriter	20 April, 2018	498,493
Anthony Bourdain	American celebrity chef, author, and travel documentarian	June 8, 2018	803,086
David Buckel	American LGBT rights lawyer and environmental activist	April 14, 2018	11,495
Brad Bufanda	American actor	November 1, 2017	3,828

Chris Cornell	American singer, songwriter, and musician	May 18, 2017	339,433
Jason Hairston	American businessman and football player	September 4, 2018	1,264
Dan Johnson	American religious leader and politician	December 13, 2017	33,070
Jill Janus	American singer	August 14, 2018	2,441
Margot Kidder	Canadian-American actress and activist	May 13, 2018	48,192
John Kivela	American politician	May 9, 2017	730
Billy Knight	American professional basketball player	July 8, 2018	102
Brian Lawler	American professional wrestler	July 29, 2018	33
Michael Mantenuto	American film actor, college ice hockey player, and Army non-commissioned officer	April 24, 2017	6,958
Jill Messick	American film producer	February 7, 2018	7,443
Tommy Page	American singer-songwriter	March 3, 2017	5,666
John Rheinecker	American professional baseball pitcher	July 18, 2017	155
Stevie Ryan	American YouTube personality, actress, and comedian	July 1, 2017	9,250
Kate Spade	American fashion designer and entrepreneur	June 5, 2018	427,067
Scott Stearney	American naval aviator and vice admiral of the United States Navy	December 1, 2018	8,068
Jean Stein	American author and editor	April 30, 2017	1,423
Jon Paul Steuer	American actor and musician	January 1, 2018	1,750
Verne Troyer	American actor, comedian, YouTuber, and stunt performer	April 21, 2018	86,867

References

1. Niederkrotenthaler T, Till B, Voracek M, Dervic K, Kapusta ND, Sonneck G. Copycat effects after media reports on suicide: A population-based ecologic study. Social Science & Medicine. 2009;69:1085–1090.
2. Sinyor M , Tran US, Garcia D, Till B, Voracek M, Niederkrotenthaler T. Suicide mortality in the United States following the suicides of Kate Spade and Anthony Bourdain. *Aust N Z J Psychiatry*. 2021;55:613-9. doi:10.1177/0004867420976844
3. Niederkrotenthaler T, Braun M, Pirkis J, et al. Association between suicide reporting in the media and suicide: systematic review and meta-analysis. *BMJ.* 2020;368:m575. doi:10.1136/bmj.m575

Supplementary Table S2. Analyses of the Lifeline Time Series, Adjusted for all of the Identified Confounding Events.

	Target event													
	Release		MTV Music Awards		Grammy awards		Video release		News impact		Logic main events		Logic main events: Excess calls n (95% CI)	Main events: % Increase (95% CI)
Effect	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	-	-
AR1	**0.79 (0.02)**	**<0.001**	**0.74 (0.03)**	**<0.001**	**0.75 (0.03)**	**<0.001**	**0.74 (0.03)**	**<0.001**	**0.73 (0.03)**	**<0.001**	**0.73 (0.03)**	**<0.001**	**-**	**-**
MA1	**1.23 (0.03)**	**<0.001**	**1.17 (0.04)**	**<0.001**	**1.17 (0.03)**	**<0.001**	**1.17 (0.03)**	**<0.001**	**1.15 (0.04)**	**<0.001**	**1.16 (0.04)**	**<0.001**	**-**	**-**
MA2	**-0.26 (0.03)**	**<0.001**	**-0.22 (0.03)**	**<0.001**	**-0.22 (0.03)**	**<0.001**	**-0.22 (0.03)**	**<0.001**	**-0.21 (0.03)**	**<0.001**	**-0.21 (0.03)**	**<0.001**	**-**	**-**
Seasonal AR1	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**-**	**-**
Seasonal MA1	**0.98 (0.005)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**-**	**-**
Target event	**205.84 (94.45)**	**0.03**	**364.82 (74.76)**	**<0.001**	**252.88 (95.00)**	**0.008**	60.27 (90.51)	0.51	158.71 (95.15)	.010	**291.62 (49.83)**	**<0.001**	**9915.08 (6594.41 to 13235.75)**	**6.87 (4.57 to 9.18)**
Exogenous events														
13 Reasons Why	**144.53 (71.46)**	**0.04**	133.08 (71.71)	0.05	134.56 (72.52)	0.06	133.15 (72.66)	0.07	127.71 (72.79)	0.08	129.28 (71.68)	0.07	11893.76 (-1031.58 to 4819.10)	3.66 (-0.32 to 7.64)
Cornell suicide	53.50 (75.56)	0.48	45.56 (74.13)	0.54	47.56 (75.02)	0.53	48.21 (75.05)	0.52	48.59 (74.81)	0.52	49.35 (73.92)	0.50	1480.50 (-2864.82 to 5827.00)	1.40 (-2.72 to 5.53)
Bennington suicide	135.02 (72.95)	0.06	77.30 (72.83)	0.29	120.81 (73.56)	0.10	119.81 (73.72)	0.10	119.79 (73.75)	0.10	84.03 (72.60)	0.25	5041.80 (-3495.96 to 13579.56)	2.19 (-1.52 to 5.91)
Avicii suicide	-96.82 (75.21)	0.20	-98.42 (73.95)	0.18	-97.55 (74.84)	0.19	-96.62 (74.88)	0.20	-94.21 (74.69)	0.21	-97.40 (73.77)	0.19	-2922.00 (-7259.68 to 1415.68)	-2.53 (-6.28 to 1.22)
Spade/Bourdain suicides	**414.28 (73.36)**	**<0.001**	**442.50 (72.85)**	**<0.001**	**430.43 (73.68)**	**<0.001**	**432.53 (73.77)**	**<0.001**	**442.91 (73.71)**	**<0.001**	**445.93 (72.74)**	**<0.001**	**28093.59 (19111.65 to 37075.53)**	**9.25 (6.29 to 12.21)**

National Suicide Prevention Week	41.47 (62.18)	.51	22.82 (61.75)	.71	42.96 (61.98)	.49	43.30 (61.99)	.49	43.08 (61.82)	.49	27.45 (61.55)	.66	384.30 (-1304.63 to 2073.23)	0.61 (-2.08 to 3.30)

Note: A SARIMA(1,1,2)(1,0,1) model was fitted to the time-series data in all analyses. Significant associations ($P < 0.05$) are printed boldface. AR# = autoregressive term with lag #; MA# = moving average term with lag #.

Supplementary Table S3. Analyses of the Suicide Time Series, Adjusted for all of the Identified Exogenous Events.

	Target event													
	Release		MTV Music Awards		Grammy awards		Video release		News impact		Logic main events		Logic main events: Excess suicides n (95% CI)	Logic main events: % Increase (95% CI)
Effect	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	-	-
AR1	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**-**	**-**
MA1	**0.87 (0.01)**	**<0.001**	**0.87 (0.01)**	**<0.001**	**0.87 (0.01)**	**<0.001**	**0.87 (0.01)**	**<0.001**	**0.87 (0.01)**	**<0.001**	**0.87 (0.01)**	**<0.001**	**-**	**-**
MA3	**0.05 (0.01)**	**<0.001**	**0.05 (0.01)**	**<0.001**	**0.05 (0.01)**	**<0.001**	**0.05 (0.01)**	**<0.001**	**0.05 (0.01)**	**<0.001**	**0.05 (0.01)**	**<0.001**	**-**	**-**
Seasonal AR1	**1.00 (0.0001)**	**<0.001**	**1.00 (0.0001)**	**<0.001**	**1.00 (0.0001)**	**<0.001**	**1.00 (0.0001)**	**<0.001**	**1.00 (0.0001)**	**<0.001**	**1.00 (0.0001)**	**<0.001**	**-**	**-**
Seasonal MA1	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**0.99 (0.003)**	**<0.001**	**-**	**-**
Target event	-11.02 (7.55)	0.15	-5.37 (3.86)	0.16	-10.12 (7.54)	0.18	7.41 (6.19)	0.23	11.76 (7.54)	0.12	**-7.20 (3.13)**	**0.02**	**-244.80 (-453.38 to -36.22)**	**-5.48 (-10.14 to -0.81)**
Exogenous events														
13 Reasons Why	5.90 (3.44)	0.09	5.73 (3.44)	0.10	5.67 (3.44)	0.10	5.66 (3.43)	0.10	5.69 (3.43)	0.10	5.90 (3.45)	0.09	542.80 (-79.30 to 1164.90)	4.37 (-0.64 to 9.38)
Cornell suicide	-0.22 (3.73)	0.95	0.14 (3.72)	0.97	0.13 (3.72)	0.97	0.14 (3.72)	0.97	0.16 (3.71)	0.97	-0.12 (3.72)	0.98	-3.60 (-222.34 to 215.14)	-0.09 (-5.41 to 5.24)
Bennington suicide	4.53 (3.50)	0.20	5.55 (3.58)	0.12	4.54 (3.51)	0.20	4.16 (3.51)	0.24	4.74 (3.49)	0.17	5.90 (3.45)	0.09	354.00 (-51.72 to 759.72)	4.33 (-0.63 to 9.30)
Avicii suicide	-1.41 (3.68)	0.70	-1.41 (3.68)	0.70	-1.41 (3.68)	0.70	-1.40 (3.67)	0.70	-1.40 (3.67)	0.70	-1.38 (3.68)	0.71	-41.40 (-257.78 to 174.98)	-1.04 (-6.47 to 4.39)
Spade/Bourdain suicides	6.34 (3.43)	0.06	6.36 (3.43)	0.06	6.35 (3.43)	0.06	6.36 (3.42)	0.06	6.36 (3.42)	0.06	6.39 (3.44)	0.06	402.57 (-22.20 to 827.34)	4.47 (-0.25 to 9.18)

National Suicide Prevention Week	-0.77 (3.92)	0.85	0.23 (3.98)	0.95	-0.77 (3.92)	0.84	-0.55 (3.93)	0.89	-0.71 (3.92)	0.86	0.51 (3.96)	0.90	7.14 (-101.52 to 115.80)	0.37 (-5.32 to 6.07)

Note: A SARIMA(1,0,3)(1,0,1) model (which included MA terms only for lags 1 and 3, but not lag 2) was fitted to the time-series data in all analyses. Significant associations ($P < 0.05$) are printed boldface. AR# = autoregressive term with lag #; MA# = moving average term with lag #.

Supplementary Table S4. Sensitivity Analysis of the Lifeline Calls Time Series, Adjusted for all of the Identified Exogenous Events: Unique calls.

	Target event													
	Release		MTV Music Awards		Grammy awards		Video release		News impact		Logic main events		Logic main events: Excess calls n (95% CI)	Logic main events: % Increase (95% CI)
Effect	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	Estimate (*SE*)	*P*	-	-
AR1	**0.73 (0.03)**	**<0.001**	**0.74 (0.02)**	**<0.001**	**0.74 (0.03)**	**<0.001**	**0.74 (0.03)**	**<0.001**	**0.73 (0.03)**	**<0.001**	**0.72 (0.03)**	**<0.001**	**-**	**-**
MA1	**1.08 (0.03)**	**<0.001**	**1.10 (0.03)**	**<0.001**	**1.09 (0.03)**	**<0.001**	**1.09 (0.03)**	**<0.001**	**1.08 (0.03)**	**<0.001**	**1.08 (0.03)**	**<0.001**	**-**	**-**
MA 2	**-0.14 (0.03)**	**<0.001**	**-0.15 (0.03)**	**<0.001**	**-0.15 (0.03)**	**<0.001**	**-0.14 (0.03)**	**<0.001**	**-0.13 (0.03)**	**<0.001**	**-0.13 (0.03)**	**<0.001**	**-**	**-**
Seasonal AR1	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**1.00 (0.001)**	**<0.001**	**-**	**-**
Seasonal MA1	**0.98 (0.004)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**0.98 (0.005)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**0.98 (0.004)**	**<0.001**	**-**	**-**
Target event	**204.11 (71.25)**	**0.004**	**314.16 (57.30)**	**<0.001**	**225.69 (71.09)**	**0.002**	37.25 (68.41)	0.59	**153.59 (71.38)**	**0.03**	**258.11 (37.71)**	**<0.001**	**8775.74 (6262.75 to 11288.73)**	**7.78 (5.55 to 10.01)**
Exogenous events														
13 Reasons Why	108.19 (55.78)	0.05	**112.55 (54.84)**	**0.04**	108.87 (55.77)	0.05	**109.93 (55.90)**	**0.049**	108.16 (55.92)	0.05	**108.29 (54.91)**	**0.049**	**9962.68 (61.31 to 19864.05)**	**4.11 (0.03 to 8.20)**
Cornell suicide	46.74 (57.42)	0.42	41.41 (56.90)	0.47	44.41 (57.46)	0.44	44.40 (57.61)	0.44	43.97 (57.46)	0.44	44.40 (56.65)	0.43	1332.00 (-1999.02 to 4663.02)	1.68 (-2.52 to 5.88)
Bennington suicide	75.10 (56.49)	0.18	39.70 (55.66)	0.48	74.32 (56.50)	0.19	75.73 (56.64)	0.19	75.13 (56.60)	0.18	43.49 (55.55)	0.43	2609.40 (-3923.28 to 9142.08)	1.48 (-2.22 to 5.18)
Avicii suicide	-109.10 (57.34)	0.06	**-112.56 (56.78)**	**0.048**	-109.01 (57.38)	0.06	-109.78 (57.53)	0.06	-108.92 (57.40)	0.06	-110.68 (56.57)	0.05	-3320.40 (-6646.72 to 5.92)	-3.70 (-7.40 to 0.01)
Spade/Bourdain suicides	**316.67 (56.48)**	**<0.001**	**316.31 (55.76)**	**<0.001**	**318.34 (56.48)**	**<0.001**	**313.57 (56.62)**	**<0.001**	**318.28 (56.56)**	**<0.001**	**321.99 (55.68)**	**<0.001**	**20285.37 (13410.00 to 27160.77)**	**8.63 (5.70 to 11.55)**

National Suicide Prevention Week	**82.09 (46.79)**	**0.008**	66.53 (46.78)	0.16	81.54 (46.75)	0.08	**80.12 (46.89)**	**0.008**	82.40 (46.81)	0.078	69.64 (46.54)	0.14	974.96 (-302.10 to 2252.02)	1.98 (-0.61 to 4.57)

Note: A SARIMA(1,1,2)(1,0,1) model (as in the main analysis) was fitted to the time-series data in all analyses. For the pre-April 7, 2017 data, this model showed a similar fit to the data as in the main analysis, with a slightly higher stationary R^2 = .71, and a still significant Box-Ljung Q = 34.65, df = 13, P = 0.001, because of slight heteroscedasticity. Significant associations ($P < 0.05$) are printed boldface. AR# = autoregressive term with lag #; MA# = moving average term with lag #.

Supplementary Table S5. Sensitivity Analyses of the Lifeline and the Suicide Time Series: Associations with All Logic Main Events, Using a Pre-Intervention Cut-Off Date of April 27, 2017, and Adjusted for all of the Identified Exogenous Events.

Effect	Lifeline time series				Suicide time series			
	Estimate (*SE*)	*P*	Excess calls n (95% CI)	% Increase (95% CI)	Estimate (*SE*)	*P*	Excess suicides n (95% CI)	% Increase (95% CI)
AR1	**0.74 (0.03)**	**<0.001**	-	-	**0.99 (0.003)**	**<0.001**	-	-
MA1	**1.17 (0.03)**	**<0.001**	-	-	**0.87 (0.01)**	**<0.001**	-	-
MA2	**-0.22 (0.03)**	**<0.001**	-		NA			
MA3	NA				**0.05 (0.01)**	**<0.001**	-	-
Seasonal AR1	**0.99 (0.001)**	**<0.001**	-	-	**1.00 (0.0001)**	**<0.001**	-	-
Seasonal MA1	**0.98 (0.004)**	**<0.001**	-	-	**0.99 (0.003)**	**<.001**	-	-
Logic main events	**288.03 (49.62)**	**<0.001**	**9793.02 (6486.34 to 13099.70)**	**6.79 (4.50 to 9.08)**	**-7.20 (3.13)**	**0.02**	**-244.80 (-453.38 to -36.22)**	**-5.48 (-10.14 to -0.81)**
Exogenous events								
13 Reasons Why	12.68 (78.74)	0.87	1166.56 (-13031.84 to 15364.96)	0.36 (-4.01 to 4.73)	5.90 (3.45)	0.09	542.80 (-79.30 to 1164.90)	4.37 (-0.64 to 9.38)
Cornell suicide	42.01 (73.43)	0.57	1260.30 (-3057.38 to 5577.98)	1.20 (-2.90 to 5.29)	-0.12 (3.72)	0.98	-3.60 (-222.34 to 215.14)	-0.09 (-5.41 to 5.24)
Bennington suicide	71.17 (71.87)	0.32	4270.20 (-4181.71 to 12722.11)	1.86 (-1.82 to 5.53)	5.93 (3.56)	0.10	355.80 (-62.86 to 774.46)	4.36 (-0.77 to 9.48)
Avicii suicide	-101.40 (73.19)	0.17	-3042.00 (-7345.57 to 1261.57)	-2.63 (-6.35 to 1.09)	-1.38 (3.68)	0.71	-41.40 (-257.78 to 174.98)	-1.04 (-6.47 to 4.39)
Spade/Bourdain suicides	**446.71 (72.02)**	**<0.001**	**28142.73 (19249.70 to 37035.76)**	**9.27 (6.34 to 12.20)**	6.39 (3.44)	0.06	402.57 (-22.20 to 827.34)	4.47 (-0.25 to 9.18)
National Suicide Prevention Week	30.98 (61.47)	0.61	433.72 (-1253.02 to 2120.46)	0.69 (-2.00 to 3.38)	0.51 (3.96)	0.90	7.14 (-101.52 to 115.80)	0.37 (-5.32 to 6.07)

Note: A SARIMA(1,1,2)(1,0,1) model was fitted to the Lifeline time-series data, whereas a SARIMA(1,0,3)(1,0,1) model (which included MA terms only for lags 1 and 3, but not lag 2) to the suicide time-series data. These models were identical to those in the main analysis, except that two further outliers were detected in the Lifeline time-series data (level shifts at Week 379, Day 7, and Week 382, Day 7). For the pre-April 27, 2017 data, stationary R^2 was .62 and .45, respectively, with Box-Ljung $Q = 42.24$, $df = 13$, $P < 0.001$ (Lifeline data), and Box-Ljung $Q = 13.84$, $df = 13$, $P = 0.39$ (suicide data). Significant associations ($P < 0.05$) are printed boldface. AR# = autoregressive term with lag #; MA# = moving average term with lag #.

Supplementary Table S6. Sensitivity Analyses of the Lifeline and the Suicide Time Series: Associations with All 5 Target Events, Adjusted for all of the Identified Exogenous Events.

Effect	Lifeline time series				Suicide time series			
	Estimate (*SE*)	*P*	Excess calls n (95% CI)	% Increase (95% CI)	Estimate (*SE*)	*P*	Excess suicides n (95% CI)	% Increase (95% CI)
AR1	**0.73 (0.03)**	**<0.001**			**0.99 (0.003)**	**<0.001**	-	-
MA1	**1.16 (0.04)**	**<0.001**			**0.87 (0.01)**	**<0.001**	-	-
MA2	**-0.21 (0.03)**	**<0.001**			NA			
MA3	NA				**0.05 (0.01)**	**<0.001**	-	-
Seasonal AR1	**0.99 (0.001)**	**<0.001**			**1.00 (0.0001)**	**<0.001**	-	-
Seasonal MA1	**0.98 (0.004)**	**<0.001**			**0.99 (0.003)**	**<0.001**	-	-
All 5 target events combined	**230.98 (40.36)**	**<0.001**	**9701.16 (6378.72 to 13023.60)**	**5.53 (3.63 to 7.42)**	-2.59 (2.80)	0.35	-108.78 (-339.28 to 121.72)	-1.94 (-6.04 to 2.17)
Exogenous events								
13 Reasons Why	130.14 (71.84)	0.07	11972.88 (-981.31 to 24927.07)	3.68 (-0.30 to 7.67)	5.75 (3.44)	0.09	529.00 (-91.30 to 1149.30)	4.26 (-0.73 to 9.25)
Cornell suicide	49.67 (74.13)	0.50	1490.10 (-2868.74 to 5848.94)	1.41 (-2.72 to 5.55)	0.06 (3.72)	0.99	1.80 (-216.94 to 220.54)	0.04 (-5.28 to 5.37)
Bennington suicide	88.42 (72.81)	0.23	5305.20 (-3257.26 to 13867.66)	2.31 (-1.42 to 6.03)	5.11 (3.56)	0.15	306.60 (-112.06 to 725.26)	3.75 (-1.37 to 8.88
Avicii suicide	-97.53 (73.98)	0.19	-2925.90 (-7275.92 to 1424.12)	-2.53 (-6.29 to 1.23)	-1.41 (3.68)	0.70	-42.30 (-258.68 to 174.08)	-1.06 (-6.49 to 4.37)
Spade/Bourdain suicides	**443.64 (72.92)**	**<0.001**	**27949.32 (18945.16 to 36953.48)**	**9.21 (6.24 to 12.17)**	6.35 (3.43)	0.06	400.05 (-23.49 to 823.59)	4.44 (-0.26 to 9.14)
National Suicide Prevention Week	33.34 (61.61)	0.59	466.76 (-1223.82 to 2157.34)	0.74 (-1.95 to 3.44)	-0.37 (3.94)	0.92	-5.18 (-113.29 to 102.93)	-0.27 (-5.94 to 5.40)

Note: A SARIMA(1,1,2)(1,0,1) model was fitted to the Lifeline time-series data, whereas a SARIMA(1,0,3)(1,0,1) model (which included MA terms only for lags 1 and 3, but not lag 2) to the suicide time-series data. Significant associations ($P < 0.05$) are printed boldface. AR# = autoregressive term with lag #; MA# = moving average term with lag #.

Supplementary Table S7. Annotated SPSS Syntax of the Statistical Models.

Syntax	Annotations
TSMODEL /MODELSUMMARY PRINT=[MODELFIT] /MODELSTATISTICS DISPLAY=YES MODELFIT=[SRSQUARE] /MODELDETAILS PRINT=[PARAMETERS] /OUTPUTFILTER DISPLAY=ALLMODELS /AUXILIARY CILEVEL=95 MAXACFLAGS=24 /MISSING USERMISSING=EXCLUDE /MODEL DEPENDENT=Lifeline_calls INDEPENDENT=Event 13RW Cornell Bennington Aviccii Spade_Bourdain NSPW PREFIX='Modell' /ARIMA AR=[1] DIFF=1 MA=[2,1] ARSEASONAL=[1] DIFFSEASONAL=0 MASEASONAL=[1] TRANSFORM=NONE CONSTANT=YES /TRANSFERFUNCTION VARIABLES=Event NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=13RW NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Cornell NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Bennington NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Aviccii NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Spade_Bourdain NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=NSPW NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /OUTLIER LOCATION=[WEEK 11 DAY 6] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 209 DAY 5] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 209 DAY 6] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 242 DAY 2] TYPE=INNOVATIONAL	Syntax for the Lifeline time series Event = dummy variable of target event of interest (this is a generic label here; in the actual data file each event has a unique denomination, which needs to be stated here) 13RW = dummy variable of 13 Reasons Why Cornell = dummy variable of Chris Cornell suicide Bennington = dummy variable of Chester Bennington suicide Aviccii = dummy variable of Avicii (Tim Bergling) suicide Spade_Bourdain = dummy variable of Kate Spade and Anthony Bourdain suicides NSPW = dummy variable of National Suicide Prevention Week For models without the above side events, delete them in the “/INDEPENDENT” line and the respective “/TRANSFERFUNCTION” lines. January 1, 2010 is Week 1, Day 6 (a Friday; the counting of days starts with Sunday = 1)

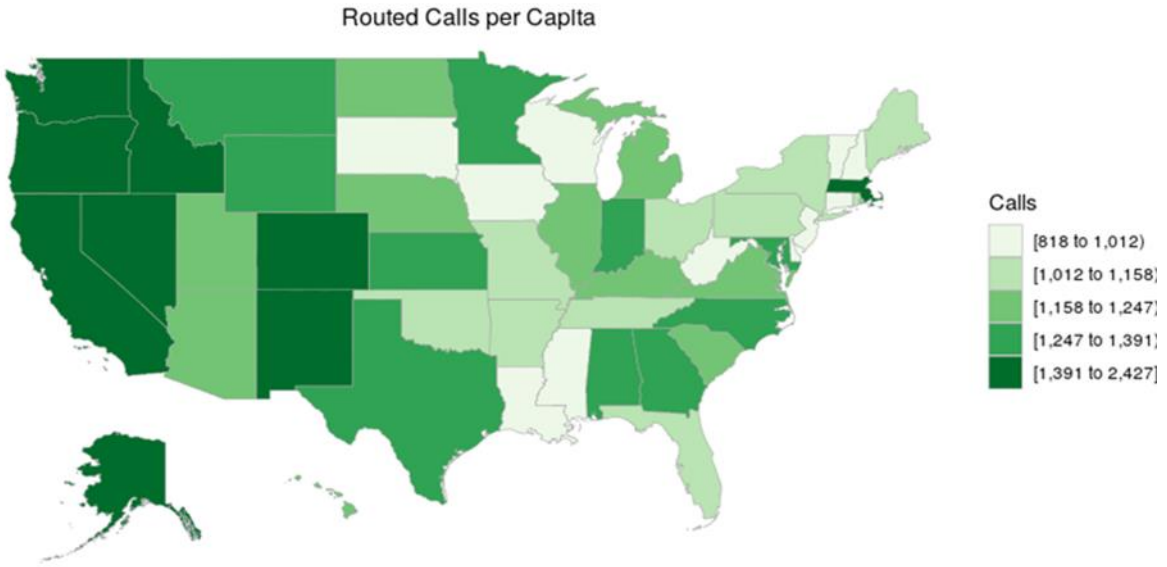
/OUTLIER LOCATION=[WEEK 242 DAY 3] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 242 DAY 4] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 256 DAY 1] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 289 DAY 3] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 289 DAY 4] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 303 DAY 2] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 303 DAY 4] TYPE=LEVELSHIFT /OUTLIER LOCATION=[WEEK 306 DAY 1] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 359 DAY 4] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 360 DAY 5] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 360 DAY 6] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 362 DAY 1] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 367 DAY 6] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 369 DAY 2] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 371 DAY 4] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 371 DAY 7] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 373 DAY 4] TYPE=INNOVATIONAL /OUTLIER LOCATION=[WEEK 373 DAY 6] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 374 DAY 1] TYPE=ADDITIVE /OUTLIER LOCATION=[WEEK 374 DAY 2] TYPE=ADDITIVE.	
TSMODEL /MODELSUMMARY PRINT=[MODELFIT] /MODELSTATISTICS DISPLAY=YES MODELFIT=[SRSQUARE] /MODELDETAILS PRINT=[PARAMETERS] /OUTPUTFILTER DISPLAY=ALLMODELS /AUXILIARY CILEVEL=95 MAXACFLAGS=24 /MISSING USERMISSING=EXCLUDE /MODEL DEPENDENT=Suicides INDEPENDENT= Event 13RW Cornell Bennington Aviccii Spade_Bourdain NSPW PREFIX='Modell' /ARIMA AR=[1] DIFF=0 MA=[3,1] ARSEASONAL=[1] DIFFSEASONAL=0 MASEASONAL=[1] TRANSFORM=NONE CONSTANT=YES /TRANSFERFUNCTION VARIABLES=Event NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0]	Syntax for the suicide time series Same as above

DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=13RW NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Cornell NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Bennington NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Aviccii NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=Spade_Bourdain NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /TRANSFERFUNCTION VARIABLES=NSPW NUM=[0] DENOM=[0] DIFF=1 NUMSEASONAL=[0] DENOMSEASONAL=[0] DIFFSEASONAL=0 DELAY=0 TRANSFORM=NONE /OUTLIER LOCATION=[WEEK 242 DAY 4] TYPE=LEVELSHIFT.	

Supplementary Text S3. Characteristics of Lifeline users: location, gender, age, primary presenting problem during call

Note: The data shown below are not representative of Lifeline users. We report them here for the purpose of highlighting that all US states and demographic groups are represented among Lifeline users. We also present information related to the primary presenting problems of a sample of callers to indicate the range of primary problems.

During the period in which Logic's song was on the charts and broadcasted widely, the Lifeline network consisted of over 160 crisis centers across the country. Calls were received from all federal states.[1] We provide a figure showing number of calls per 100,000 residents for each US state for the period from April 1 2017 to December 31, 2018 (covering the entire post-release observation period). Calls were well distributed according to population density, although there was some evidence that calls were more commonly originating from the West Coast of the United States.



National Suicide Prevention Lifeline, 2021.

Data on gender and age are not routinely collected during calls and also not available on a daily level from any of the Lifeline centers. There is only aggregated information from some specific centers, so called back-up centers.

Data from these centers are not a random sample of all Lifeline users. Back-up centers respond to calls that local centers are not answering or covering geographically. This means that there is likely some gender and age skew in who is more likely to wait longer to have their call answered; and some states route more callers to backup centers than others. The proportion of missing values is relatively large. We report demographics for calls answered in the period April 2017 to December 2018 below. In total, during this period, n=703,694 total calls were received in 9 centers that provided data. Additionally, we report data from September 2017 only (data from the same 9 centers, n=38,094 total calls).

Consistent with international literature,[2] the current gender and age data suggest that all age and gender groups are represented in service users. Adults up to 64 years of age, who make the largest

proportion of suicide deaths, appear overrepresented, as are girls and women. There are no major discrepancies in age and gender of service users in September 2017, the month when attention to the Logic song was most pronounced, as compared to the overall post-release observation period starting with April 2017 and ending December 2018.

Age	**N (%), April 2017 to December 2018**	**N (%), September 2017**	**Gender**	**N (%),April 2017 to December 2018**	**N (%), September 2017**
5-14	16,475 (3.4)	845 (3.6)	Female	249,925 (51.7)	12,817 (54.7)
15-24	102,469 (21.2)	5,826 (24.9)	Male	192,924 (39.9)	9,662 (41.2)
25-34	75,835 (15.7)	3,622 (15.4)	Other	366 (0.2)	31 (0.1)
35-44	49,375 (10.2)	2,092 (8.9)	Transgender	852 (0.3)	40 (0.2)
45-54	42,856 (8.9)	1,864 (8.0)	Unknown	39,544 (8.2)	897 (3.8)
55-64	31,880 (6.6)	1,404 (6.0)			
65-74	16,522 (3.4)	607 (2.9)			
75-84	3,976 (0.8)	169 (0.7)			
85+	1,663 (0.3)	30 (0.1)			
Unknown	142,470 (29.5)	6,915 (29.5)			
Total	**483,521**	**23,347**		**483,611**	**23,447**

With regard to the primary presenting problem of callers, we similarly report data from the period April 2017 to December 2018 and data from September 2017 only. These data indicate that for all but approximately 4% of calls (*n* = 24,002) in the post-release period, a specific primary presenting problem was identified. The proportion of callers with the presenting problem "suicide" (which includes users calling because of their suicidal thoughts and feelings or suicidal behaviour) was 28.9% (*n* = 162,986) in the post-release period April 2017 to December 2018, whereas 28.2% (*n* = 774) during September 2017, the period with strongest attention to the Logic song. These data indicate that there was no noted discrepancy in terms of the primary problems presented and registered at these centers during September 2017 compared to the overall period after the release (April 2017 to December 2018).

Primary presenting problem	**April 2017 to December 2018 N (%)**	**September 2017**
Abuse / victimization	9,030 (1.6)	428 (1.5)
Family	113,118 (20.0)	5,304 (18.9)
Financial/ basic needs	9,031 (1.6)	418 (1.5)
Government agency	420 (0.07)	18 (0.06)
Job loss	9,065 (1.6)	447 (1.6)
Legal problems	1,757 (0.3)	77 (0.28)
Lifeline complaints	643 (0.1)	33 (0.1)
Medical	14,131 (2.5)	667 (2.4)
Mental Health	150,270 (26.6)	7,914 (28.3)
Other	47,779 (8.4)	2,901 (10.4)
Sexuality / LGBTQ	3,463 (0.6)	168 (0.6)
Substance abuse / addiction	12,432 (2.2)	576 (2.1)

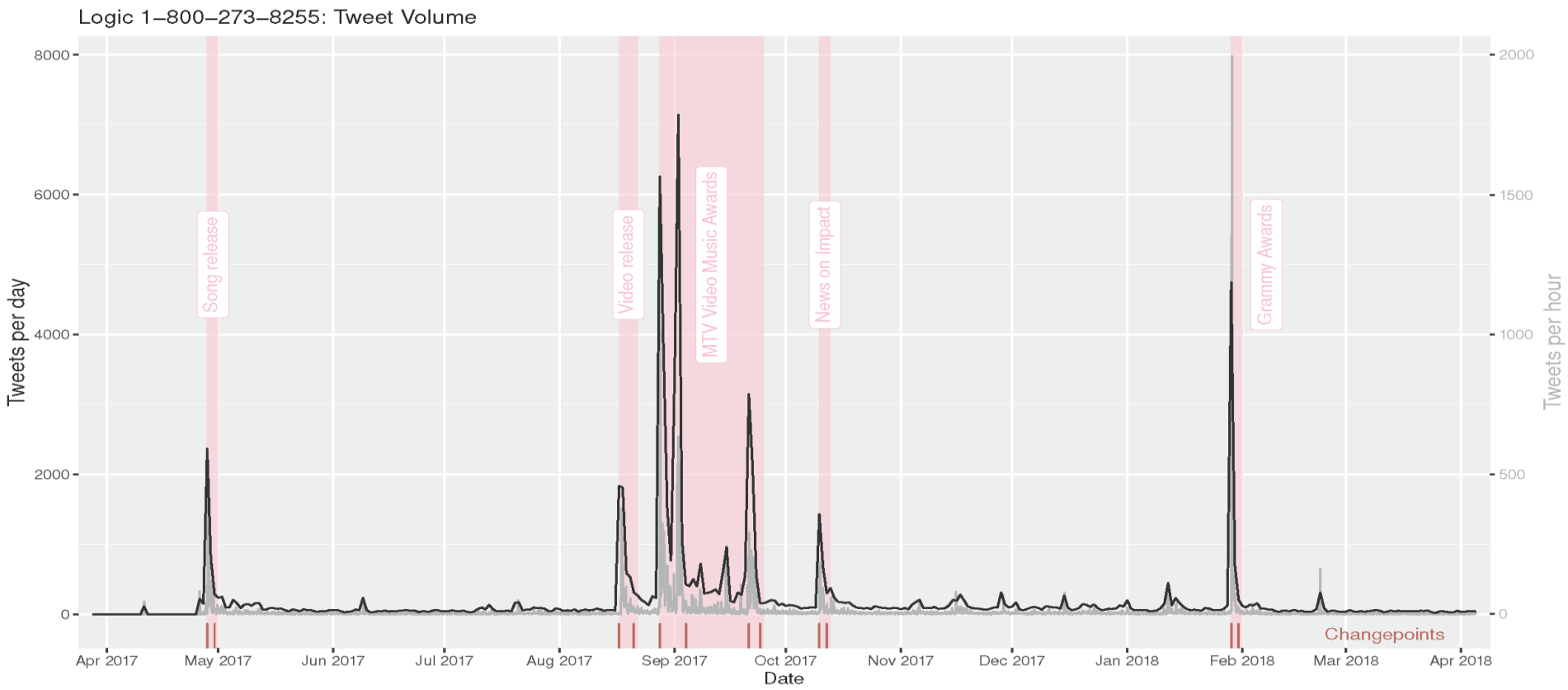
Suicide (including suicidal thoughts and feelings)	162,986 (28.9)	7,886 (28.2)
VA / Active Military related	6,689 (1.2)	381 (1.4)
Unknown	24,002 (4.2)	774 (2.8)
Total	**564,816**	**27,992**

References

1. National Suicide Prevention Lifeline. *Lifeline Backup Center Demographic Report* 1/1/2017 - 12/31/2017 (internal report).
2. World Health Organization. Preventing suicide: a resource for establishing a crisis line. Geneva: World Health Organization; 2018.

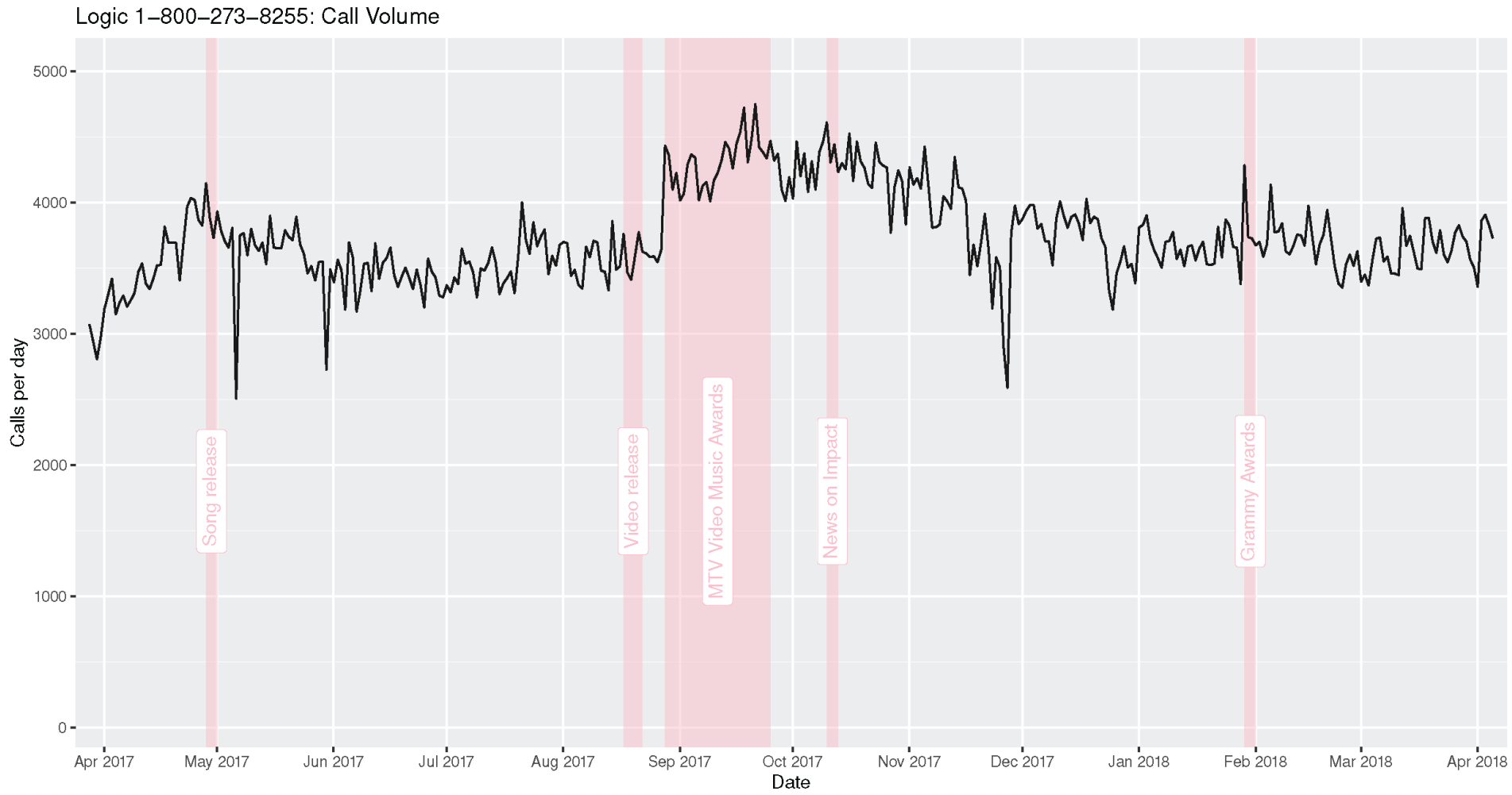
Supplementary Figure S1. Daily number of (A) Tweets including Logic and 1-800-273-8255, (B) calls to the National Suicide Prevention Lifeline, and (C) suicides -- United States, April 2017 to April 2018

(A)

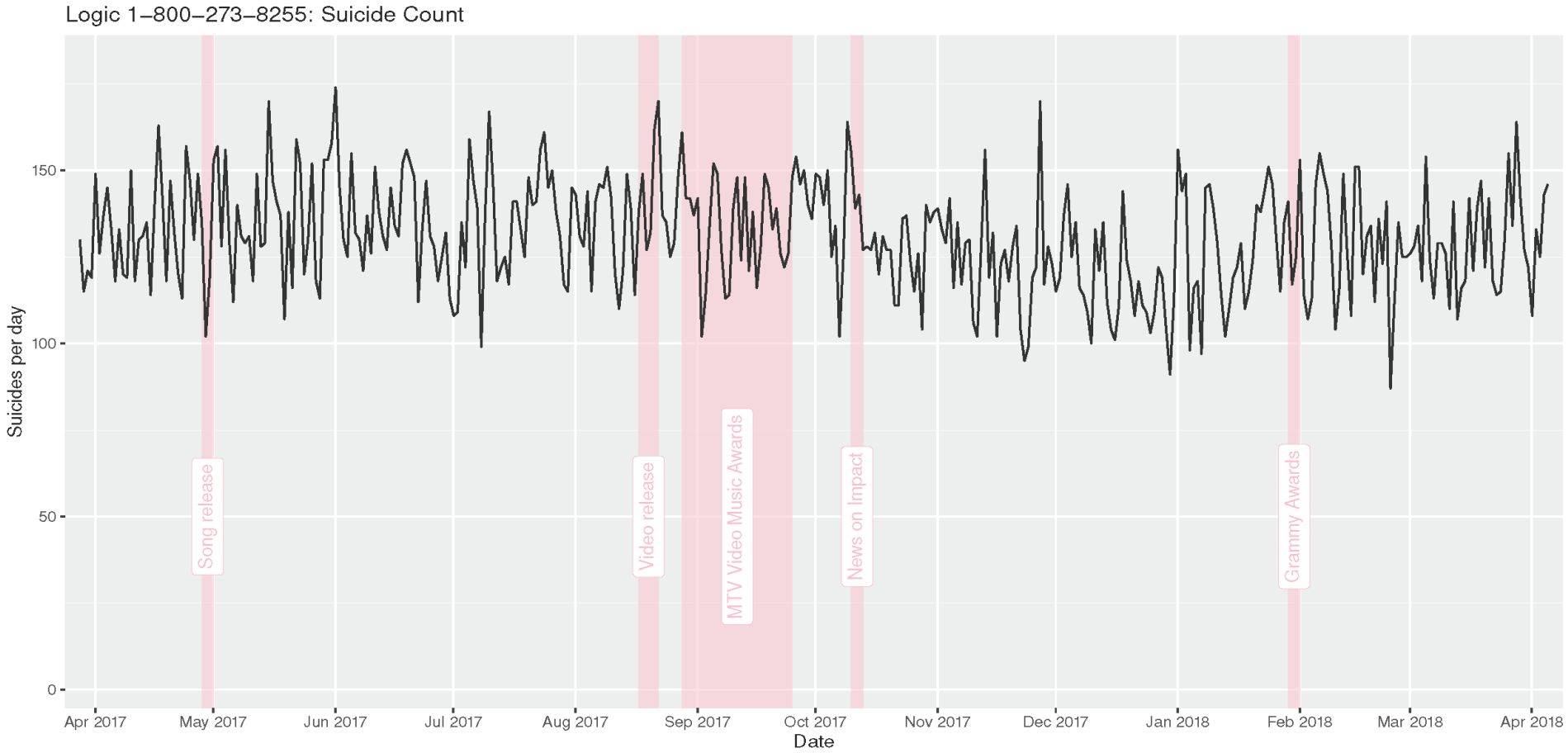


Coloured areas represent the duration of impact, as identified through visual inspection and used in the analysis. The short vertical lines at the bottom (below the x-axis) are the dates based on the findings from the change-point analysis (for details, see Appendix, Supplemental Text S2, p. 5).

(B)


Logic 1–800–273–8255: Call Volume
Calls per day
5000
4000
3000
2000
1000
0
Song release
Video release
MTV Video Music Awards
News on Impact
Grammy Awards
Apr 2017
May 2017
Jun 2017
Jul 2017
Aug 2017
Sep 2017
Oct 2017
Nov 2017
Dec 2017
Jan 2018
Feb 2018
Mar 2018
Apr 2018
Date

(C)



Source of Suicide Data: Centers for Disease Control and Prevention; suicide defined by International Classification of Diseases, 10th Revision (ICD-10) underlying cause of death codes X60–X84, Y87.0, and U03.